

TRANSPORT PHENOMENA

ChE 319 – SPRING 2025

Unique # - 16010

Meets in CPE 2.216 on MWF from 1:00 p.m. – 2:00 p.m.

Instructor - Prof. Venkat Ganesan, CPE 3.414

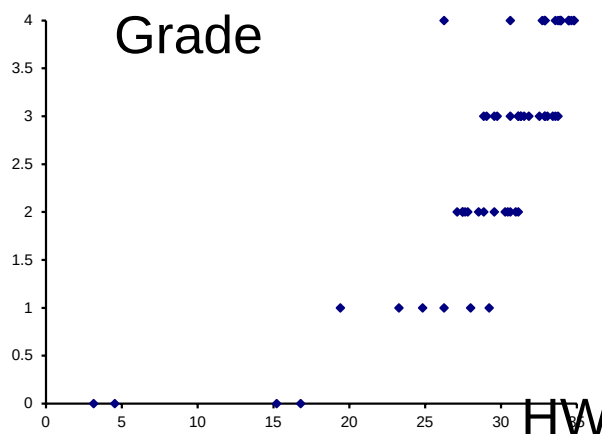
- TEXTBOOK:** R. B. Bird, W. E. Stewart and E. N. Lightfoot, *Transport Phenomena*, 2nd Ed., John Wiley & Sons, Inc., 2002.
- PREREQUISITES:** ChE 317-Introduction to Chemical Engineering Analysis and M 427J Differential Equations with Linear Algebra
- RECITATION SESSIONS:** Wednesday 4:00 - 6:00 p.m. CPE 2.216.
- OFFICE HOURS:** Monday 2:00 – 3:30 pm – CPE 3.414.
- TEACHING ASSISTANT:** Collin Farrell (Office Hrs: Monday: 3:30 pm – 5 pm in CPE 2.222; Tuesday: 2:30 pm – 4pm in CPE 4.446)
- TUTORS:** Alex Lau (Office Hrs: Monday: 6 pm – 7:30 pm in CPE 2.222); Adit Trivedi (Office Hrs: Tuesday: 6 pm – 7:30 pm in CPE 2.222).
- HOMEWORK POLICY:** There will be a homework assignment for nearly every week during the semester. *Homeworks will not be collected, however, you will have a closed book/closed notes quiz over the homework set that is scheduled for that day during the recitation.* These recitation quizzes will be composed of problems very similar to homework problems (with some simple modifications) and/or verbatim homework problems. The recitation quiz grades will contribute 25% to your final grade. **Recitation quizzes cannot be made up, however your single lowest recitation quiz score will be dropped.** If more than one recitation quiz is missed the additional missed quizzes will be scored as a zero. Solution keys to HWs will not be posted. However, the solution to recitation quizzes will be posted online.

You may work together with your classmates on homework but don't deny yourself the opportunity to "struggle" with difficult concepts and problems (by having friends "spoon-feed" you). It is my opinion that one learns a great deal in this endeavor....homework problems are your means of "practicing" engineering just as an athlete practices to improve his/her athletic ability.

Many of the assigned homework problems will be very challenging....indeed, the subject of this course involves rich and powerful intellectual machinery for solving tremendously important technological problems. Most of you will have to work very hard to become competent in Transport Phenomena...as has everyone that I know. One of the common myths of students is that "they are not talented enough or smart enough" to learn this subject well (in addition to others)....I believe that each of you are plenty bright enough to master this subject beyond your wildest expectations if you have the desire and will to work at it very hard.

This is an extremely important course in the Chemical Engineering Curriculum. Many of my colleagues that are practicing engineers cite a course such as this one as a key element in the development of their own engineering skills. Many of you will learn a very new way to think about describing physical phenomena....one of the important components of the course is to attain the ability to "derive," based on engineering balances, a differential equation that describes a situation of interest and to solve this equation for use in design or understanding of engineering equipment.

The following was the correlation between the HW scores and final grades from the last time I taught it (the quizzes were weighted at 35% then):



As you can see, 98% of students securing < 27/35 obtained a grade of D or below. To reinforce this, your scores in the first and the second recitation quiz will be closely monitored. Since these quizzes are at most just variants of your HWs, **you will be expected to secure a score of 80% or above** in both these quizzes.

Scholastic dishonesty will not be tolerated and incidents of dishonesty will be reported to the Dean of Student Affairs.

TEACHING PHILOSOPHY:

There are several important concepts and techniques in this course that I will not be able to discuss in great detail in lecture. However, I plan to assign homework problems and generate test questions that will deal with some or all of these concepts. You will be expected to learn about these techniques outside of class from reading the textbook or library research. Further, study of the appropriate sections of the text prior to lecture enhances the learning experience for me as well as you since our interaction can be much more of a two-way dialogue rather than the typical "I speak, you listen" discourse. Such advanced preparation also allows us to rapidly cover the rudiments so that we may address the finer points of the subject more thoroughly.

One definition of a chemical engineering education is "the study of the quantitative analysis of problems of chemical separations, chemical reaction engineering, heat and momentum transport, thermodynamics and molecular engineering." Thus, we will discuss these subjects and the advanced mathematics required to address them as well the physical origin

of transport phenomena so that we might develop a physical intuition regarding the subject. We will also discuss several novel solutions to some engineering problems that hopefully will motivate you to think in creative ways.

LECTURE ATTENDANCE:

It is important and expected that you will attend all lectures. I will try to do my part to make the lectures interesting and worth attending. Please be prompt since late arrivals tend to distract others in the class. There is a strong correlation between class attendance and participation in course activities and understanding and performance in the course, therefore, I strongly advise you to attend lectures. **If you miss class or arrive late and miss receiving a handout please do not disturb me in my office later.....get the materials you missed from one of your classmates.**

Please be courteous to others during class: don't sit in class talking about topics that are not included in the course. It is acceptable to briefly and quietly discuss some element of the lecture with a classmate sitting nearby.

Use of cell-phones is prohibited during the lecture. Students who are repeat offenders will be penalized points from their final scores.

To ensure that there is a consistent pattern of attendance, there will be a number (almost every lecture) of very short lecture quizzes. These will be held in the beginning of the lecture and will typically not require any preparation and take very little time (< 5 mins). Sometimes, these will be based on a problem, I may post as "pre-lecture" problem. These quizzes will only be graded on 1 or 0 scale. You will be allowed to drop the two lowest quiz scores.

EXAMS:

Midterm and final exams will be very challenging and you will need to study extensively to perform well. These exams will be closed book and closed notes. **You will be allowed one "formula card" made up of a 3 in. x 5 in. card** with anything that you like written on it (both sides may be used). **Exams cannot be made up.....**if a Midterm Exam must be missed due to illness or family crisis then the Final Exam score will be substituted for the missed exam. This can only be done once and students that must miss more than one Midterm Exam should see the Deans Office (ECJ 2.200) regarding dropping the course. If you have questions about the manner in which your exam was graded then write a note describing the discrepancy, attach the note to your exam and return the note and exam to me and I will review this together with the grader. This procedure must be followed within one class period of the time the graded exams are initially returned otherwise we will not re-grade your test. In addition, we will review the entire exam to look for any other discrepancies (positive or negative).

GRADING:

We will strive to fairly grade all of your quizzes and exams. Since you will have had the problems/templates for the recitation quizzes we expect nearly perfect work ... if it is clear that you have little clue about how to work the problem you will not receive any partial credit for that problem. Additionally, there are a few items for which you will be heavily penalized:

(i) serious conceptual algebra, calculus, and/or differential equations errors, (ii) conceptual transport errors (i.e., wrong coordinate system, etc.), and (iii) errors in units (even for problems that are entirely composed of variables).

GRADING:

We will strive to fairly grade all of your quizzes and exams. Since you will have had the problems/templates for the recitation quizzes we expect nearly perfect work ... if it is clear that you have little clue about how to work the problem you will not receive any partial credit for that problem. The grades will range from A – F (including + and -).

The grades will be curved and the typical range for the grades are the following (G: score; x : average; σ : SD):

$G > x + \sigma$: A (typically around 90)

$x < G < x + \sigma$: B, B^+, A^- (typically around 75-90)

$x - \sigma < G < x$: C^-, C, C^+, B^- (typically around 65 – 75)

$G < x - \sigma$: $D \dots F$ (typically below 65)

EVALUATION:

Recitation Quizzes	25%	- Drop lowest rec. quiz grade.*
Lecture Quizzes	5%	- Drop two lowest lec. quiz grades.*

Midterm Exams	16.66% each (3 exams for a total of 50%) [#]
Final Exam	<u>20%</u>
	100% - Possible Final Percentage

*If you must miss a recitation quiz then that will be the quiz grade that you miss will drop. Additional missed quizzes will be scored zero.

&Missed lecture quizzes result in a grade of zero for that quiz.

[#]You may substitute the grade you earn on the Final Exam for the lowest score you earn on a Midterm Exam (provided all three Midterm Exams are attempted and your Final Exam grade is higher than at least one of your Midterm Exam grades).

The University of Texas at Austin provides, upon request, appropriate academic adjustments for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-4241 TDD or the College of Engineering Director of Students with Disabilities at 471-4382.

COURSE CONTENT:

Heat Transfer:

Conduction - 1-Dim., Transient, Rectangular Coord., Cylindrical Coord.
Radiation - Simple Concepts

Convection - Laminar Flow, Turbulent Flow, Detailed analysis and balances, Use of Heat Transfer Correlation's, Development of Heat Transfer Correlation's

Fluid Mech.:

Macroscopic balances, microscopic balances.
Laminar and turbulent flow.

Mass Transfer:

Ficks law, simple diffusion problems, mass transfer correlations

Lecture dates in which I will be absent:

February 12, February 26, February 28, March 3.

To make up for these 4 missed lectures, I will be teaching 75 mins on most Wednesdays (indicated by * below) during recitation quiz time in the plan. Two of these 75 minute lectures will compensate for 1 missed lecture. On those dates, the recitation quizzes will be held between 1 – 1:50 pm in CPE 2.216.

Plan For Recitations

Jan. 15 – Math Review

Jan. 22* - Recitation Quiz over HW#1. 75 minute Lecture at 4pm.

Jan. 29* - Recitation Quiz over HW#2. 75 minute Lecture at 4pm.

Feb. 5* - Recitation Quiz over Hw#3. 75 minute Lecture at 4pm.

Feb. 12 - Recitation Quiz over Hw#4. No lecture.

Feb. 19 – Mid-Term Exam 1.

Feb. 26 - Recitation Quiz over Hw#5. No lecture.

Feb. 28 – No lecture.

March 3 – No lecture.

March 5* – Recitation Quiz over Hw#6. 75 minute Lecture at 4pm.

March 12* – Recitation Quiz over Hw#7. 75 minute Lecture at 4pm.

March 19 – Spring break 😊

March 26 - Midterm Exam #2

April 2* - Recitation Quiz over Hw#8. 75 minute Lecture at 4pm.

April 9* - Recitation Quiz over Hw#9. 75 minute Lecture at 4pm

April 16* - Recitation Quiz over Hw#10. 75 minute Lecture at 4pm.

April 23 – Mid-Term 3

April 28 - Last day of this class.

**(TBD) - Final Exam for this class.

USEFUL BOOKS IN THE LIBRARY:

Engineering Library

J. P. Holman, *Heat Transfer*, 5th Ed., QC 320, H64, 1981.

J. P. Holman, *Heat Transfer*, 6th Ed., QC 320, H64, 1986.

S. Whitaker, *Introduction to Fluid Mechanics*, QA 911, W38.

M. N. Ozisik, *Heat Transfer: A Basic Approach*, TJ 260, O96, 1984.

R. B. Bird, et al., *Transport Phenomena*, QA 929, B5, 1960.

H.S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, QC 321, C28, 1959.

W. M. Kays, *Convective Heat and Mass Transfer*, 2nd Ed., QC 327, K37, 1980.

W. M. Kays and M. E. Crawford, *Convective Heat and Mass Transfer*, 3rd Ed., QC 327, K37, 1993.

E. Kreyszig, *Advanced Engineering Mathematics*, 7th Ed., QA 401, K7, 1992.

W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, 5th Ed., QA 371, B773, 1992.

J. R. Welty, C. E. Wicks and R. E. Wilson, *Fundamentals of Momentum, Heat and Mass Transfer*, 3rd Ed., TA 357, W45, 1984.

J. R. Welty, C. E. Wicks and R. E. Wilson, *Fundamentals of Momentum, Heat and Mass Transfer*, 2nd Ed., TA 357, W45, 1976.